

Tawfik Yasser Tawfik

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EXPERIENCE

- onebank, Data Engineer (Full-Time)** (May 2026 - Present)
- Apache Nifi, Cloudera, SQL, Tableau, ETL, Pipelines.
- Egypt University of Informatics, Assistant Lecturer (Full-Time)** (Dec 2025 - May 2026)
- Distributed Systems, Data Engineering, and Big Data Systems.
- Nile University, Teaching Assistant (Full-Time), ITCS, AI Program** (Sep 2022 - Sep 2024)
- Restructured "Introduction to Big Data" syllabus to include hands-on labs with **Docker**, **Hadoop**, **Apache Spark**, and **NoSQL** tools; increased practical lab completion and student project readiness.
 - [Organized](#) academic events (e.g., [NILES2023](#)) and co-authored research papers; supervised student projects on cloud-based clusters using **GCP**.
- Re-Hub, Data Engineer (Contract: Remote)** (Apr 2022 - Jun 2022)
- Designed and implemented daily **ETL pipelines** in **Python** and **PostgreSQL**, deployed as scheduled DAGs in **Apache Airflow**; improved data quality by **88%**
 - Integrated external sources via **REST APIs** (Airtable), implemented robust parsing and extraction logic for large datasets and reduced manual data-cleaning tasks.
 - Wrote unit checks and monitoring tasks in Airflow to detect schema drift and prevent bad-data propagation; documented workflows and handover procedures for the production team.

EDUCATION

- MSc in Informatics (Big Data) - Nile University, Cairo, Egypt (GPA: 3.92) (Sep 2022 - May 2024)
 - Thesis: Keyed Watermarks: A Fine-grained Watermark Generation for **Apache Flink**. ([Link](#))
 - Set up and managed cloud-based clusters on Google Cloud Platform (GCP) for data engineering and machine-learning workflows, and federated learning applications.
- BSc in Computer Science - Cairo University, Cairo, Egypt (GPA: 3.01) (Sep 2018 - Jul 2022)

PROJECTS

- Executed ETL pipelines using **Flink**, **Kafka**, **PostgreSQL**, **Elasticsearch**, and **Kibana**, encompassing container setup using **Docker**, data generation, and consumption. Enhanced with streamlined pipeline development, database integration, and multi-sink output for categorized financial transactions, enabling comprehensive insights via **Kibana** dashboards. ([Link](#))
- Developed a comprehensive data pipeline utilizing **Reddit API**, **Airflow**, **Celery**, **Postgres**, **S3**, **AWS Glue**, **Athena**, and **Redshift** for efficient ETL processes. The solution seamlessly extracts, transforms, and loads Reddit data into Redshift for advanced analytics and querying capabilities. ([Link](#))
- Developed a project showcasing real-time data streaming using TCP Socket, **Apache Spark**, **Kafka**, **Elasticsearch**, and **Kibana**. Guided through data acquisition, processing, sentiment analysis with TextBlob, and seamless integration with Kafka, **Elasticsearch**, and **Kibana** for robust pipelines. ([Link](#))
- Developed a scalable data engineering solution leveraging AWS services to ingest, transform, and analyze YouTube video data, ensuring security and efficiency while utilizing **AWS IAM**, **Amazon S3**, **AWS Glue**, **Lambda**, and **QuickSight** for centralized storage, ETL, and reporting. ([Link](#))
- Streamlined **CDC** (Change Data Capture) processes using **Debezium**, **Kafka**, and **Postgres**, achieving seamless schema evolution, comprehensive change tracking with user attribution, and precise decimal handling for dependable data synchronization and insights. ([Link](#))

SKILLS

- Python, SQL, PostgreSQL, Spark, PySpark, Flink, Kafka, Airflow, Docker, AWS (S3, Lambda, Glue, Redshift), GCP, Elasticsearch, Kibana, ETL, Data Modeling, Parquet, Delta Lake, Git, Linux.

PUBLICATIONS

- Keyed Watermarks: A Fine-grained Watermark Generation for Apache Flink. ([link](#))
- Keyed Watermarks: A Fine-grained Tracking of Event-time in Apache Flink. ([link](#))
- Image Captioning Using Vision Encoder Decoder Model. ([link](#))
- Advanced Network Malware Detection: Integrating PySpark and ML Techniques. ([link](#))
- Sentiment Analysis for Arabic Product Reviews Using LLMs and Knowledge Graphs. ([link](#))